Quantum Oracle Forensics

1. The problem

We are given a sealed oracle $U|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$ on 8 input bits, with m = 1 or m = 8 output bits, and at most 60 queries (oracle gates × shots). We must name which of eight families f belongs to and report its secret exactly: constant (c), balanced, linear (a, b), Grover (marked input t), randfun, simon (s), simon2 (a 2-dimensional subspace H) or perm. Every query is noisy: each oracle call misfires with probability 4% and each measured bit flips with probability 2%. The stress test doubles both. A task scores only if the family and every bit of the secret are right.

2. How we approached it

We analyzed the given problem statement before writing any code:

- **Usage of "m".** The solver is told m, so the task splits into a 5-way problem (m = 1) and a 3-way problem (m = 8) at zero cost.
- **Each family matches a known quantum algorithm, but none covers all eight.** Deutsch–Jozsa / Bernstein–Vazirani reads the Walsh spectrum of f, Grover search finds a marked input, Simon sampling reveals a hidden subspace, and classical queries read values such as b and c. The real task is choosing which experiment to run next.
- **Some pairs are hard by design.** Deutsch–Jozsa cannot tell constant-0 from Grover (P(z = 0) is 1 vs 0.984) or see b and c. Balanced and randfun differ mainly in P(z = 0): exactly 0 vs about 0.16. Simon, simon2 and perm all use the same circuit.
- **Noise makes every decision statistical.** About 15% of 8-bit readouts contain an error, and a 10-call Grover circuit runs cleanly only about two-thirds of the time. No single shot can be trusted.

We first built hand-designed decision rules and tested them in simulation. That work found three weaknesses: decisions resting on one or two readings, single bit flips silently corrupting answers, and fixed thresholds that would break under doubled noise. It also showed that the textbook number of Grover iterations (12) is not optimal under misfires. We checked designs against "Quantum Computation & Quantum Information by Nielsen & Chuang" and tested before adopting anything. This led to the final design: one statistical engine that handles every family the same way.

3. The solution

3.1 Core method

- **Exact Bayesian inference.** Every possible answer is a hypothesis: 2 constant, 510 linear (a, b), 256 Grover t, balanced and randfun (with a prior over its density) for m = 1, and perm, 255 simon s and all ten 795 two-dimensional subspaces for m = 8. After each shot, each hypothesis is re-weighted by the exact probability of what was measured, with the noise model built in.
- **Noise level is not assumed.** Every hypothesis is paired with both nominal (4%, 2%) and doubled (8%, 4%) noise, and the posterior averages over both. The same code therefore handles the stress test without re-tuning.
- **Adaptive experiments.** Before each shot, the solver picks the circuit with the highest expected information gain about the answer per query spent.
- **Sequential stopping.** It answers as soon as the best answer has posterior probability $\geq 1 - \delta$, with $\delta = 0.003$ chosen by a sweep over 0.01, 0.003 and 0.001. Easy oracles finish in a few queries, and hard ones get the budget they need.
- **Readout repetition (our extension).** Spare qubits receive CNOT copies of the measured qubits just before measurement. Readout flips hit each copy independently, so agreeing copies are almost surely right and disagreements become known erasures instead of silent errors. This costs no queries.

3.2 Circuits

Circuit	Queries	What it reveals
Deutsch–Jozsa / Bernstein–Vazirani	1	constant and Grover → z = 0; linear → z = a exactly; balanced never gives z = 0; randfun gives z = 0 w.p. $(1 - 2w)^2$

Circuit	Queries	What it reveals
Grover echo (2 or 4 iterations)	2–4	an even number of iterations returns a balanced function exactly to z = 0 but not a randfun: P(z = 0) is 0.86 vs 0.09–0.35
Grover search (6–12 iterations)	6–12	the marked input t; 8–10 iterations give the best success per query under misfires
Simon sampling (H·U·H)	1	z orthogonal to the hidden subspace: one vector for simon, two for simon2, none for perm
Classical query of f(x)	1	b for linear, c for constant, and verification of a Grover candidate

The circuit menu. For m = 8 only Simon sampling is needed; one exact likelihood scores all 11 051 hypotheses after every shot.

Typical paths chosen by the solver: linear takes 2–3 Deutsch–Jozsa shots to fix "a" and 1–2 classical reads to fix b (about 5 queries in total). Grover alternates Grover searches with classical checks of the best candidate. balanced and randfun mix echo and Deutsch–Jozsa shots (about 24). The m = 8 families take 23–33 Simon shots.

4. Why a quantum approach, and how well it fits

The challenge uses the query model: the oracle is given, and the only cost is the number of calls. This is exactly the setting where quantum algorithms have provable advantages, and the eight families line up with the classic examples.

- **Search (Grover vs constant-0).** Deciding whether any input is marked is the OR problem. A randomized classical algorithm needs Θ(N) queries, about 128 on average to find t among 256, which does not fit in 60. A quantum algorithm needs Θ(√N) (Nielsen & Chuang Ch. 6.6–6.7), about 12 iterations. This is the one place where quantum is unavoidable at this size.
- **Hidden subspaces (simon, simon2, perm).** These are hidden-subgroup problems over 8-bit strings with XOR. Classically you must wait for two inputs to collide on the same output, which is rare for simon and perm within 60 queries. One Simon shot gives a linear constraint on the hidden subspace almost every time.
- **Global properties (linear, balanced, randfun).** Deutsch–Jozsa reads the whole Walsh spectrum in one query, so "a" in a linear function appears in a single shot instead of about 8 classical reads. Here the advantage is real but modest, since random classical sampling also works, just with twice or more queries.

We measured this directly with a classical-only baseline that uses the same Bayesian engine but no superposition. Classical alone is enough for constant, linear, balanced, randfun and simon2, at higher cost. It fails on Grover, simon and perm. The quantum circuits are therefore used where they are necessary and where they save queries, and plain classical reads are kept where they are the cheapest tool (b, c, and verifying t).

Family	Quantum	Classical only	Queries, ours	Queries, classical
Grover	1.00	0.20	23.5	56.1
simon	1.00	0.16	29.1	60.0
perm	1.00	0.22	32.7	60.0
linear	1.00	1.00	5.2	20.5
balanced	1.00	0.98	25.0	57.3
randfun	0.98	0.90	23.8	53.4
simon2	1.00	0.98	22.9	60.0
constant	1.00	1.00	33.9	36.9

Accuracy and mean queries per family, quantum vs classical-only, same 50 oracles per family, nominal noise.

5. How accurate the results are

All results come from a fixed-seed benchmark: 50 random oracles per family (400 tasks), with the same oracles in every experiment, so the comparisons are paired.

Experiment	Accuracy	Worst family	Query saving	Mean queries
Our solver, nominal noise	**0.998**	**0.980**	**0.591**	**24.5**
Our solver, stress noise	**0.990**	**0.960**	**0.463**	**32.2**
Budget cap 20 / 30 / 40 / 50	0.917 / 0.983 / 0.993 / 0.998	0.68 / 0.92 / 0.96 / 0.98	0.73 / 0.65 / 0.62 / 0.60	16.7 / 21.1 / 22.8 / 23.8

Experiment	Accuracy	Worst family	Query saving	Mean queries
Without readout copies, stress	0.960	0.860	0.400	36.9
Classical only	0.680	0.160	0.232	50.5
Random guess	0.113	0.000	n/a	0

Main results and section-9 baselines. Query saving = 1 − queries/60, averaged over correct tasks.

- **Accuracy.** 399 of 400 tasks are correct at nominal noise (95% confidence interval 98.6–99.96%), and 396 of 400 under stress (97.5–99.6%).
- **Robustness.** Every family is at least 98% accurate at nominal noise and at least 96% under stress. The weakest case is Grover under stress, where the doubled misfire rate breaks more Grover searches.
- **Efficiency.** The solver uses on average 24.5 of the 60 queries, stopping as soon as it is sure. Linear needs about 5 queries. Constant-0 always uses the full budget: proving that no input is marked is the OR problem above, and stopping earlier would cost Grover accuracy.
- **Failures.** One error at nominal noise (a randfun that produced data about 1000:1 in favour of constant-1). Four under stress (two Grover reported as constant-0, one balanced as randfun, one randfun as constant-1).

6. How we checked it

- The local simulator reproduces every reference value in the problem statement and Qiskit's bit order. It agrees with Qiskit's own Statevector simulation to within 10^{-9}.
- A second, independently written simulator agrees with the first to within 10^{-13} on every circuit type, including 20-qubit Simon circuits with misfires.
- A second, independently written solver reaches the same accuracy on the same oracles (1.000 nominal, 0.983 stress) but uses about 4 more queries per task. This confirms both the results and that our design is the more efficient one.

7. Limitations

- For balanced and randfun, only P(z = 0) is modelled exactly; other outcomes are treated as uniform. A small uniform floor in every likelihood limits the effect of model error.
- Noise is averaged over two levels (nominal and doubled). A different, un-modelled noise level or correlated errors would be handled only approximately.
- Our circuits use up to 20 qubits and 7-control gates.

8. Conclusion

The problem is a sequence of decisions under noise, and quantum query algorithms supply the most informative experiments for those decisions. Grover search and Simon sampling are necessary at this size; Deutsch–Jozsa and the Grover echo are not strictly necessary but cut the query count by half. Combining them with exact Bayesian inference, adaptive experiment choice and free readout repetition gives 99.8% accuracy using 41% of the budget on average, and 99.0% accuracy under doubled noise with no re-tuning.